Calcium-40

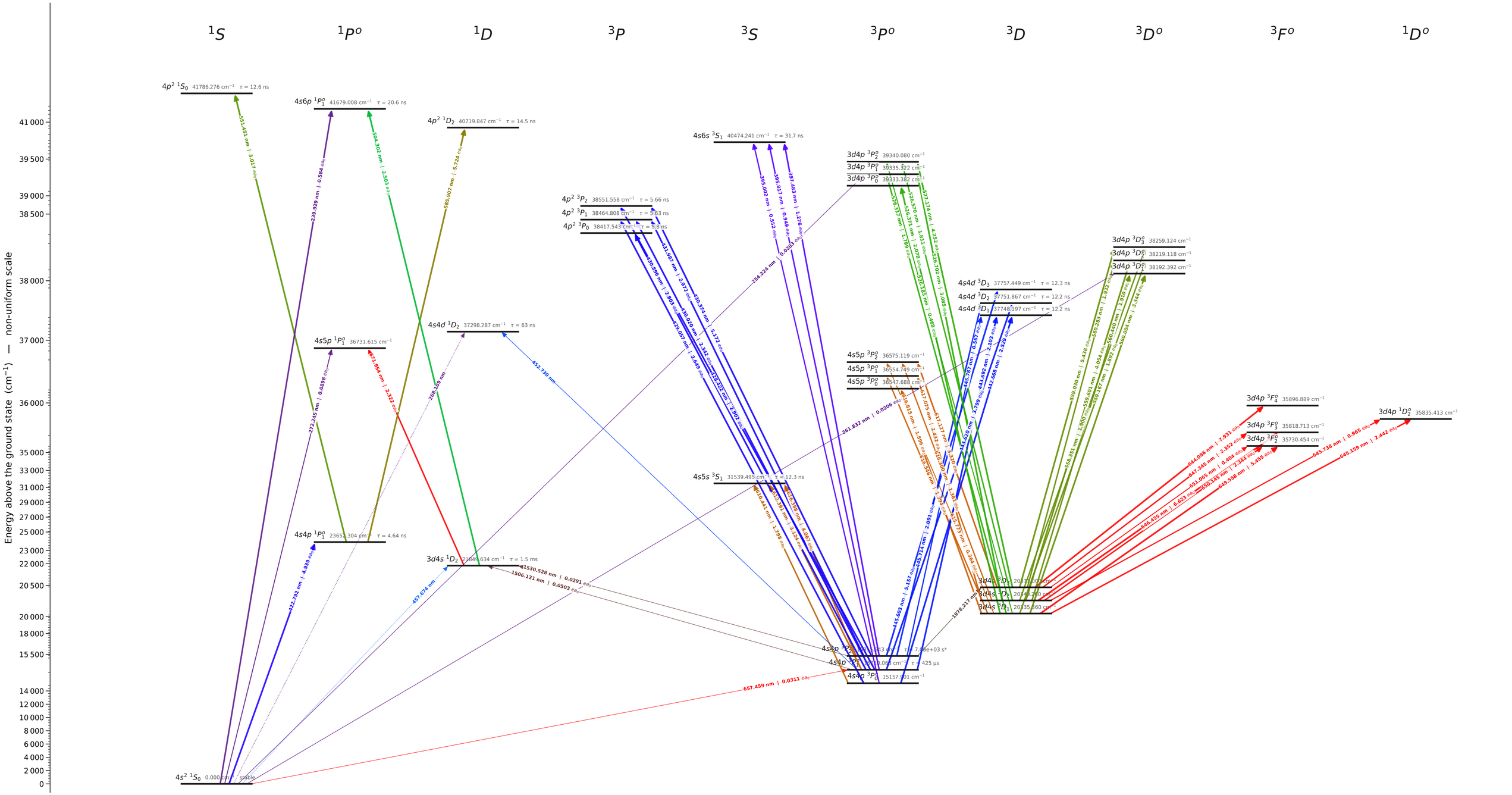
^{40}Ca (Ca I), nuclear spin $I = 0$. Energy levels and transitions below 2 µm: 35 levels, 61 lines.

Ca$^+$ ionisation limit 49305.92 cm^{-1} = 6.11315 eV (202.815 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	\|⟨J‖er‖J'⟩\| (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$4s^2\ ^1S_0$ – $4s4d\ ^1D_2$	268.1088	268.0292	1118.1745	–	–	2.526	–	two-photon transition
$4s^2\ ^1S_0$ – $4s5p\ ^1P_1^o$	272.2450	272.1644	1101.1861	0.090	2.700e+05	–	–	272.16 nm line
$4s^2\ ^1S_0$ – $4s4p\ ^1P_1^o$	422.7917	422.6727	709.078373	4.939	2.180e+08	34.31	100	422.67 nm
$4s4p\ ^3P_2^o$ – $4p^2\ ^3P_2$	430.3738	430.2528	696.5862	5.172	1.360e+08	28.12	77	430.252 nm
$4s4p\ ^3P_1^o$ – $4s4d\ ^1D_2$	452.7299	452.6030	662.1883	–	–	2.526	–	452.7 nm quench line used to broaden the 657 nm transitio…
$4s^2\ ^1S_0$ – $3d4s\ ^1D_2$	457.6736	457.5453	655.0355	–	4.000e+01	0.0001061	6	457.5 nm electric-quadrupole line
$4s4p\ ^3P_2^o$ – $4s5s\ ^3S_1$	616.3878	616.2173	486.3698	4.067	4.770e+07	12.94	58.7	616.2 nm line of the 4s4p 3P - 4s5s 3S1 triplet
$4s^2\ ^1S_0$ – $4s4p\ ^3P_1^o$	657.4594	657.2779	455.986240	0.031	2.300e+03	0.0003745	97.8	657.28 nm
$3d4s\ ^1D_2$ – $4s5p\ ^1P_1^o$	671.9536	671.7681	446.1506	2.322	1.200e+07	–	–	671.77 nm
$4s4p\ ^3P_1^o$ – $3d4s\ ^1D_2$	1506.1214	1505.7099	199.0493	0.050	3.000e+02	0.0001061	83	decay of the 1D2 reservoir back to the ground state via 3…
$4s4p\ ^3P_2^o$ – $3d4s\ ^1D_2$	1530.5285	1530.1103	195.8751	0.029	9.600e+01	0.0001061	17	loss channel of the 423 nm MOT into metastable 3P2
$4s4p\ ^3P_2^o$ – $3d4s\ ^3D_3$	1978.2171	1977.6771	151.5468	–	–	–	–	1978 nm closed transition of the metastable

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$4s^2\ ^1S_0$ – $4s4d\ ^1D_2$	268.109	42-40	1268.7(3)
$4s^2\ ^1S_0$ – $4s4d\ ^1D_2$	268.109	43-40	1895.5(4)
$4s^2\ ^1S_0$ – $4s4d\ ^1D_2$	268.109	44-40	2445.8(4)
$4s^2\ ^1S_0$ – $4s4d\ ^1D_2$	268.109	46-40	3561.0(9)
$4s^2\ ^1S_0$ – $4s4d\ ^1D_2$	268.109	48-40	4582.3(8)
$4s^2\ ^1S_0$ – $4s5p\ ^1P_1^o$	272.245	42-40	967(9)
$4s^2\ ^1S_0$ – $4s5p\ ^1P_1^o$	272.245	43-40	1455(9)
$4s^2\ ^1S_0$ – $4s5p\ ^1P_1^o$	272.245	44-40	1879(14)
$4s^2\ ^1S_0$ – $4s5p\ ^1P_1^o$	272.245	46-40	2746(16)
$4s^2\ ^1S_0$ – $4s5p\ ^1P_1^o$	272.245	48-40	3528(16)
$4s^2\ ^1S_0$ – $4s4p\ ^1P_1^o$	422.792	42-40	393.2(3)
$4s^2\ ^1S_0$ – $4s4p\ ^1P_1^o$	422.792	43-40	611.1(5)
$4s^2\ ^1S_0$ – $4s4p\ ^1P_1^o$	422.792	44-40	773.79(19)
$4s^2\ ^1S_0$ – $4s4p\ ^1P_1^o$	422.792	46-40	1159.5(8)
$4s^2\ ^1S_0$ – $4s4p\ ^1P_1^o$	422.792	48-40	1513.0(4)
$4s4p\ ^3P_2^o$ – $4s5s\ ^3S_1$	616.388	43-40	-38.2(20)
$4s4p\ ^3P_2^o$ – $4s5s\ ^3S_1$	616.388	48-40	-116(4)
$4s^2\ ^1S_0$ – $4s4p\ ^3P_1^o$	657.459	42-40	509.5(3)
$4s^2\ ^1S_0$ – $4s4p\ ^3P_1^o$	657.459	43-40	782.2(7)
$4s^2\ ^1S_0$ – $4s4p\ ^3P_1^o$	657.459	44-40	996.2(5)
$4s^2\ ^1S_0$ – $4s4p\ ^3P_1^o$	657.459	46-40	1481.1(7)
$4s^2\ ^1S_0$ – $4s4p\ ^3P_1^o$	657.459	48-40	1922.5(8)

Shift = ν(first isotope) − ν(second isotope).